

Ethical Hacking Practical Master Guide

(Experiments 1–15)

Designed for quick use during practical exams. Follow steps, capture screenshots, show expected output.

Exp 1 Metadata2Go

Main Motive: Find digital footprints from metadata.

Install/Check: Install: None. Open metadata2go.com.

Steps / Commands:

Upload file → View Metadata.

Screenshot To Take: Screenshot: Metadata page showing Author/Created Date.

What To Point Out: Show examiner: Author, created date, modified date, software used.

Exp 2 FTK Imager

Main Motive: View digital evidence safely.

Install/Check: Install FTK Imager (Windows).

Steps / Commands:

Create C:\Evidence\secret.txt → File > Add Evidence Item > Folder.

Screenshot To Take: Screenshot: Evidence Tree + file contents.

What To Point Out: Show examiner: timestamps, properties, file preview.

Exp 3 Autopsy

Main Motive: Perform forensic investigation.

Install/Check: Install Autopsy.

Steps / Commands:

New Case → New Host → Logical Files → Add Evidence folder → Enable Hash Lookup.

Screenshot To Take: Screenshot: Data Sources + Results.

What To Point Out: Show examiner: files analyzed and report generation.

Exp 4 Nessus

Main Motive: Find vulnerabilities.

Install/Check: Install Nessus Essentials.

Steps / Commands:

Open localhost:8834 → New Scan → Basic Scan → Target 127.0.0.1 → Launch.

Screenshot To Take: Screenshot: Vulnerability results page.

What To Point Out: Show examiner: Critical/High/Medium findings.

Exp 5 John the Ripper

Main Motive: Crack password protected ZIP.

Install/Check: Check: `john --version`

Steps / Commands:

```
echo 'This is experiment 5' > sc.txt
```

```
zip -e sc.zip sc.txt
```

```
zip2john sc.zip > ziphash.txt
```

```
john --wordlist=/usr/share/wordlists/rockyou.txt ziphash.txt
```

Screenshot To Take: Screenshot: Password cracked output.

What To Point Out: Show examiner: recovered password.

Exp 6 Hydra

Main Motive: Online password attack.

Install/Check: Check SSH running: `sudo systemctl start ssh`

Steps / Commands:

```
echo '123456' > pass.txt
```

```
echo 'password' >> pass.txt
```

```
echo 'kali' >> pass.txt
```

```
hydra -v -l kali -P pass.txt ssh://127.0.0.1
```

Screenshot To Take: Screenshot: login and password found.

What To Point Out: Show examiner: valid credentials discovered.

Exp 7 Metasploit

Main Motive: Privilege escalation analysis.

Install/Check: Metasploit already in Kali.

Steps / Commands:

```
msfconsole
```

```
search local_exploit_suggester
```

```
use post/multi/recon/local_exploit_suggester
```

```
show options
```

Screenshot To Take: Screenshot: SESSION required.

What To Point Out: Show examiner: module loaded, explain vertical/horizontal escalation.

Exp 8 Steghide

Main Motive: Hide data inside image.

Install/Check: `sudo apt install steghide -y`

Steps / Commands:

```
echo 'My Secret Message' > secret.txt
```

```
steghide embed -cf image.jpg -ef secret.txt
```

```
steghide extract -sf image.jpg
```

Screenshot To Take: Screenshot: embedding done + extraction done.

What To Point Out: Show examiner: hidden file successfully recovered.

Exp 9 SET Toolkit

Main Motive: Understand phishing/social engineering.

Install/Check: SET already installed.

Steps / Commands:

sudo setoolkit

1 > 5 > 1 > 1

Screenshot To Take: Screenshot: Mass Mailer Attack menu.

What To Point Out: Show examiner: explain phishing workflow. No need to send email.

Exp 10 Nmap Stealth Scan

Main Motive: Perform stealth reconnaissance.

Install/Check: sudo apt install nmap -y if needed.

Steps / Commands:

nmap -sT 127.0.0.1

nmap -sS 127.0.0.1

nmap -sS -T2 127.0.0.1

nmap -f 127.0.0.1

nmap -D RND:5 127.0.0.1

Screenshot To Take: Screenshot: open ports.

What To Point Out: Show examiner: difference between -sT and -sS.

Exp 11 Session Hijacking

Main Motive: Analyze session cookies.

Install/Check: sudo apt install wireshark ettercap-graphical -y

Steps / Commands:

sudo wireshark → Start capture → Filter http

Screenshot To Take: Screenshot: Cookie / Set-Cookie packets.

What To Point Out: Show examiner: session identifiers visible.

Exp 12 DoS/DDoS Analysis

Main Motive: Analyze attack traffic.

Install/Check: Wireshark required.

Steps / Commands:

Start capture → Filters tcp / icmp.

Screenshot To Take: Screenshot: large packet traffic.

What To Point Out: Show examiner: DoS vs DDoS observations.

Exp 13 Shodan/Nmap

Main Motive: IoT reconnaissance.

Install/Check: Nmap installed.

Steps / Commands:

```
nmap -sn 127.0.0.1  
nmap -sV 127.0.0.1  
nmap -O 127.0.0.1  
nmap -A 127.0.0.1  
nmap --script vuln 127.0.0.1
```

Screenshot To Take: Screenshot: OS/service detection.

What To Point Out: Show examiner: host, service and OS detected.

Exp 14 OpenSSL/GPG

Main Motive: Encrypt and decrypt files.

Install/Check: openssl version
gpg --version

Steps / Commands:

```
echo 'Hello' > secret.txt  
openssl enc -aes-256-cbc -salt -in secret.txt -out secret.enc  
openssl enc -aes-256-cbc -d -in secret.enc -out decrypted.txt  
gpg -c secret.txt  
gpg -d secret.txt.gpg
```

Screenshot To Take: Screenshot: decrypted file output.

What To Point Out: Show examiner: original text recovered.

Exp 15 Hashcat

Main Motive: Recover password from hash.

Install/Check: hashcat --version

Steps / Commands:

```
echo -n 'hello' | md5sum  
hashcat -m 0 -a 0 hash.txt words.txt  
hashcat -m 0 --show hash.txt
```

Screenshot To Take: Screenshot: Status Cracked.

What To Point Out: Show examiner: recovered password 'hello'.